

Cardio AI: Explainable Heart Disease Prediction using CNN-LSTM Attention Network with Grad-CAM Interpretability

Project Proposal for UCS503P
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1 Higher-order goal

... secondary framing

This project aims to improve *clinical transparency in cardiac diagnostics* by making AI-driven heart disease predictions interpretable to both clinicians and patients. While the broader aspiration is trustworthy medical AI, the immediate engineering objective is to build a deployable, explainable prediction system that translates deep learning confidence scores into actionable clinical narratives.

2 Time-to-value

... primary heuristic

- We will deliver meaningful increments quickly by first training and validating the CNN-LSTM-Attention model on the Cleveland dataset, then layering explainability (Grad-CAM, Attention, SHAP) and the clinical web interface in successive iterations.
- We will prioritize fast feedback over perfection by using short iteration cycles: model accuracy is validated before explainability modules are integrated, and each module is independently testable.
- Success is defined by what can be measured in the first iteration (holdout AUC ≥ 0.85), then improved based on cross-validation and clinical-alignment feedback.

3 Problem Statement

Cardiovascular disease remains the leading cause of mortality globally. Early prediction from routine clinical measurements (blood pressure, cholesterol, ECG markers) can significantly improve patient outcomes. However, current challenges include:

- **Black-box predictions:** Deep learning models achieve high accuracy but offer no clinical reasoning, making physicians reluctant to trust automated diagnoses.

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- **Feature opacity:** With 13 heterogeneous clinical features (numerical vitals, categorical ECG types, binary indicators), understanding *which* features drive a specific patient’s risk is non-trivial.
- **Limited visual evidence:** Clinicians lack visual artifacts (heatmaps, feature importance charts) that map model confidence back to physiological parameters.
- **No longitudinal tracking:** Existing tools rarely support patient-history-aware predictions or trend analysis across repeated assessments.

Why this matters now (contextual relevance): As AI-assisted diagnosis tools enter clinical workflows, regulatory bodies and medical ethics demand interpretable, auditable predictions — not just accurate ones.

4 Proposed Solution

... Software Proposition

4.1 Overview

Build a **web-based** explainable cardiac risk prediction system with the following components:

- **CNN-LSTM-Attention deep learning model:** trained on the UCI Cleveland Heart Disease dataset (303 records, 13 clinical features) with GAF image encoding for spatiotemporal feature extraction.
- **Multi-method explainability engine:** Grad-CAM heatmaps, attention weight visualization, and SHAP values providing three complementary lenses on model reasoning.
- **Clinical report generator:** automated PDF reports with narrative summaries, feature-by-feature clinical interpretation, and guideline-referenced evaluations.
- **Patient management portal:** Flask-based web interface with patient registration, longitudinal history tracking, and searchable records via MySQL backend.
- **Interactive visualization dashboard:** Plotly-based charts for real-time XAI comparison (Attention vs. Grad-CAM vs. SHAP).

4.2 Core workflow

1. Clinician enters patient clinical parameters (age, blood pressure, cholesterol, ECG markers, etc.) via the web form.
2. System preprocesses inputs using a `ClinicalRangeScaler` (physiological-bound normalization), converts to GAF image representation, and runs prediction through the CNN-LSTM-Attention model.
3. Explainability engine generates Grad-CAM heatmap, extracts attention weights, and computes SHAP attributions.
4. System produces an interactive dashboard with XAI visualizations and a downloadable PDF clinical report with guideline-referenced narrative.
5. Patient record is persisted to MySQL database for longitudinal tracking and future comparisons.

4.3 Operational constraints for an educational setting

- Use the publicly available Cleveland dataset (303 records) — no patient consent or privacy concerns.
- Keep the web interface responsive and usable on standard hardware without GPU requirements at inference time.
- Design for local deployment (Flask + MySQL) with a clear path to containerized hosting.

5 Solution Approach

... Engineering focus

This project combines deep learning engineering with clinical interpretability, focusing on robust pipeline design and multi-method XAI rather than novel architecture research.

5.1 Data preprocessing pipeline

... non-research

- **Missing value handling:** median imputation for Cleveland `ca` and `thal` columns (marked as `?`).
- **Outlier treatment:** IQR-based capping to preserve dataset size on a small ($n = 303$) dataset.
- **Feature scaling:** custom `ClinicalRangeScaler` using fixed physiological boundaries (e.g., age: 25–85, BP: 80–200 mmHg) instead of dataset-dependent statistics, ensuring out-of-sample consistency.
- **Encoding:** one-hot encoding for categorical features (chest pain type, `thal`, slope, etc.) with explicit feature-to-index mapping for XAI aggregation.
- **Image transformation:** Gramian Angular Summation Field (GASF) encoding converts 1D feature vectors to 2D images for Conv2D input.
- **Class balancing:** SMOTE oversampling on the training split with fallback to class-weighted loss.

5.2 Model architecture

... deep learning

A hybrid CNN-LSTM-Attention architecture:

- **Conv2D layers:** extract spatial correlations from GAF-encoded feature images.
- **LSTM layer:** captures sequential/temporal dependencies across feature dimensions.
- **Attention mechanism:** learns to weight feature contributions, providing an inherently interpretable bottleneck.
- **Output:** sigmoid activation for binary classification (heart disease present/absent).

5.3 Explainability methods

... interpretable AI

Three complementary XAI techniques:

- **Grad-CAM:** gradient-weighted class activation maps on the final Conv2D layer, averaged to 1D feature importance.
- **Attention weights:** extracted from the attention layer and aggregated from one-hot indices back to original clinical features.

- **SHAP values:** model-agnostic Shapley attributions providing per-feature contribution with directionality.

5.4 Web architecture

... web-based solution

A 3-tier architecture:

- **Frontend:** HTML/CSS/JS with Plotly.js for interactive XAI charts and responsive patient input forms.
- **Backend API:** Flask (Python) serving prediction, explainability, PDF generation, and patient search endpoints.
- **Database:** MySQL storing patients, clinical records, AI diagnoses, and PDF report paths.

6 Evaluation Criterion

... measurable and attributable

We define success using measurable metrics tied to both model performance and clinical interpretability.

6.1 Primary evaluation metric

Area Under the ROC Curve (AUC-ROC): measures the model’s ability to discriminate between heart disease and normal cases across all classification thresholds.

- Target: $AUC \geq 0.85$ on the 20% holdout test set.
- Attribution: computed from model probability outputs against ground-truth binary labels.

6.2 Secondary evaluation metrics

- **Accuracy, Precision, Recall, F1-Score:** standard classification metrics on the holdout test set.
- **Cross-validation stability:** 5-fold stratified CV to verify generalization (target: CV AUC std ≤ 0.05).
- **XAI clinical alignment:** top-3 features identified by Grad-CAM, Attention, and SHAP are cross-referenced against established clinical risk factors (e.g., `ca`, `thal`, `cp`, `oldpeak`) per ACC/AHA guidelines.
- **Report completeness:** generated PDF reports contain narrative summary, feature-level evaluation against clinical guidelines, and risk-stratified recommendations.

6.3 Pilot validation plan

... high-level

- Validate predictions on all 303 Cleveland dataset records using nearest-neighbour ground-truth matching.
- Demonstrate the web portal end-to-end: form submission $\rightarrow$ prediction $\rightarrow$ XAI dashboard $\rightarrow$ PDF report $\rightarrow$ patient history retrieval.
- Collect qualitative feedback from 3–5 peers on report readability and XAI visualization clarity.

7 Scalability

... with foresight

The proposition should scale in both theoretical and practical senses.

1. **Scalable (theoretically):** The system should support growth in patient volume and model complexity by:
 - decoupling the prediction engine from the web server (API-first design),
 - stateless API design supporting future horizontal scaling,
 - using SQL queries for patient history search.
2. **Operationally deployable (practically):** The system should run on standard infrastructure:
 - Flask + Docker-hosted MySQL on any Windows/Linux/macOS host,
 - containerizable via Docker for reproducible deployment,
 - model weights serialized for inference without retraining.

8 Engine availability heuristic

A mature set of “engines” (tooling/libraries) is available to implement this project:

- **TensorFlow/Keras:** CNN-LSTM-Attention model construction, training, Grad-CAM computation.
- **scikit-learn:** preprocessing, train-test splitting, classification metrics, cross-validation.
- **imbalanced-learn:** SMOTE for class balancing.
- **SHAP:** model-agnostic Shapley value computation.
- **Flask:** lightweight web framework for REST API and template rendering.
- **PyMySQL:** MySQL database connectivity.
- **Plotly.js:** interactive client-side XAI visualizations.
- **Microsoft Edge (headless):** HTML-to-PDF clinical report conversion.
- **Pandas, NumPy, Matplotlib, Seaborn:** data manipulation, EDA, and static plotting.

We will use these mature components to focus on pipeline correctness, explainability quality, and clinical alignment rather than inventing new infrastructure.

9 Project scope and deliverables

... iteration-friendly

9.1 Initial deliverable

... first iteration

- EDA pipeline: missing values, duplicates, class distribution, correlation analysis
- Preprocessing pipeline: ClinicalRangeScaler, one-hot encoding, GASF transformation, SMOTE
- CNN-LSTM-Attention model: training with holdout evaluation and 5-fold cross-validation
- Grad-CAM and Attention weight extraction with one-hot-to-original feature aggregation
- Basic Flask endpoint: accepts patient features, returns prediction and XAI weights

9.2 Subsequent deliverables

- SHAP integration for model-agnostic feature attribution
- Interactive Plotly.js dashboard for XAI comparison (Attention vs. Grad-CAM vs. SHAP)
- PDF clinical report generator with guideline-referenced narrative summaries
- Patient management: registration, longitudinal history tracking, search API
- GAF image visualization and 2D Grad-CAM heatmap display
- Subgroup-aware performance metrics (by age group, sex)

10 Risks and mitigations

- **Small dataset** ($n = 303$): mitigate via stratified splitting, SMOTE, cross-validation, and IQR outlier capping to maximize data utility.
- **Overfitting on small data:** EarlyStopping, ReduceLROnPlateau, and ModelCheckpoint callbacks; dropout regularization in the network.
- **XAI disagreement across methods:** present all three methods (Grad-CAM, Attention, SHAP) side-by-side; highlight consensus features rather than relying on a single method.
- **Clinical misinterpretation:** clearly label all outputs as “AI-assisted, not diagnostic”; reference ACC/AHA clinical guidelines in reports.
- **TensorFlow deployment issues:** maintain a lightweight fallback inference path using NumPy-based forward pass for environments without GPU/native TF support.

11 Summary

This project proposes a deployable, explainable heart disease prediction system combining a CNN-LSTM-Attention deep learning model with multi-method interpretability (Grad-CAM, Attention weights, SHAP). It meets the course criteria by emphasizing time-to-value through iterative model-explainability-interface delivery, using measurable evaluation criteria (AUC-ROC ≥ 0.85 , cross-validation stability, clinical feature alignment), and producing clinician-readable PDF reports. The system is designed for scalable deployment via a Flask web portal with MySQL-backed patient history management.